COSC 120 Final Project Planning Worksheet

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The final project takes the place of the final exam. You may work **individually or with one partner**. Your goal is to design, build, test, and present an extended Python program that brings together the major ideas from the course.

1 What Do You Want to Build?

Describe a program that would be useful, interesting, scientific, creative, or fun. It should solve a problem, model something, organize information, automate a task, or create an interactive experience.

Working title: _____
Partner, if any: _____
What will the program do?

Who would use it, and why?

2 Inputs, Outputs, and Data

What information will enter the program, and what results will the program produce?

Which data structures seem useful?

☐ Lists
☐ Dictionaries
☐ Nested data structures
☐ Objects created from classes

Give one example of data your program will store.

3 Organize the Program

Your project should make meaningful use of several ideas from the course, including functions, conditionals and loops, lists and/or dictionaries, and at least one class with useful attributes and methods. Use modules and imports where appropriate. Sketch a possible organization below:

```
main.py
   |
   +-- module1.py
   +-- module2.py
```

What is an example of a class the program could use? What would the class store, and what would it do?

4 Think About an Algorithm

An algorithm is a sequence of steps for solving a problem. Identify one part of your project where the program must search, sort, compare, simulate, calculate, update, or make a decision repeatedly. How could you organize the steps so the program does not perform unnecessary work?

Could a better data structure or algorithm make this part more efficient? Explain briefly.

5 Define a Realistic First Version

What is the **minimum working version** of your project? That is, the simplest version that solves the problem?

If that version works, what feature would you add next?

6 Plan the Presentation

At the end of the semester, you will present your project to the class. Your presentation should demonstrate the program and explain how it works. Plan to show:

- the problem or goal,
- a live or recorded demonstration,
- the important classes, functions, and data structures,
- one interesting algorithm or design decision,
- what worked, what was difficult, and what you learned.

Before leaving today: Be prepared to describe your project idea in one or two sentences and identify the first piece of code you would write.